

Computational Mechanics:

1. Overview & Scope

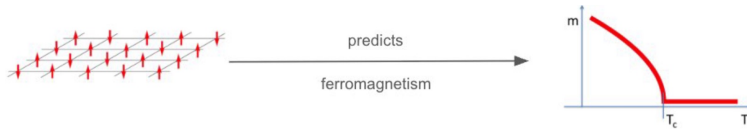
ILIAD INTENSIVE - SUMMER 2026

Outline:

- 1) What is Computational Mechanics?
- 2) Computational Mechanics & AI Safety
- 3) Overview — where is the program at?
- 4) Scope — what will we tackle today?

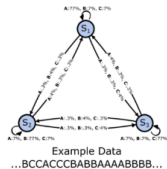
What is Computational Mechanics?

* Statistical mechanics makes predictions about the behaviour of systems by considering their statistical properties. E.g.



Figures from
Yuvan et al.

* Computational mechanics (comp-mech) makes predictions about the behaviour of systems by considering their computational properties. E.g.



Figures from
Shai et al.

What is Computational Mechanics?

Some questions that are central to comp-mech:

- * How much of the past does a system store?
- * How does the system store this information?
- * How does this information affect system behaviour?

We will consider each of these questions for transformers today!

What is Computational Mechanics?

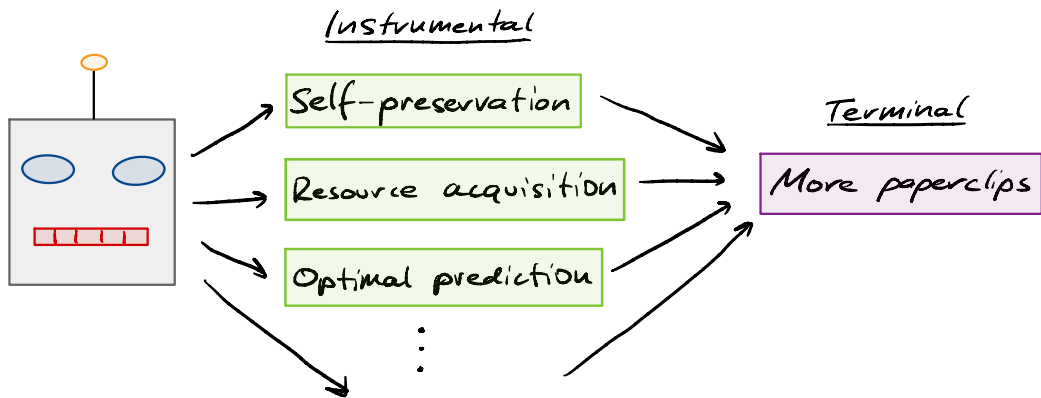
Not all systems are interesting! Comp-mech places an emphasis on those performing optimal prediction. In this setting the central question becomes:

What are convergent structures relevant for optimal prediction?

Insight on this question greatly constrains our search space when considering optimal predictors.

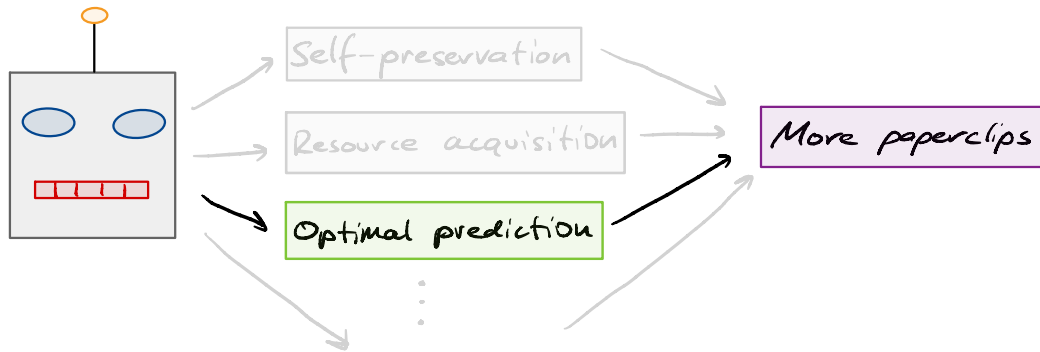
Computational Mechanics & AI Safety

We have strong reason to believe that AI systems will pursue a range of instrumental goals in service of their terminal goal



Computational Mechanics & AI Safety

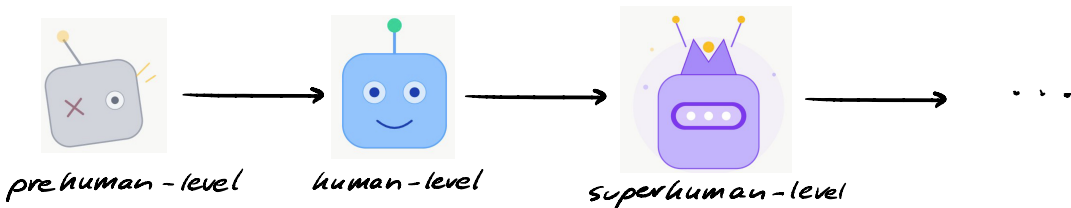
Comp-mech uses an instrumental goal as an intervention target point



By considering internal representations consistent with optimal prediction the search space is reduced.

Computational Mechanics & AI Safety

Targeting an instrumental goal is scale-free – it will be relevant to current and future models



Which contrasts scale-sensitive approaches e.g. AI control that are only applicable up to a fixed capability scale.

Moreover, as capabilities increase models will better approximate optimal predictors.

Overview - where is the program at?

Comp-mech applied to AI safety is a young research program (~ 2 years). There is much to do!

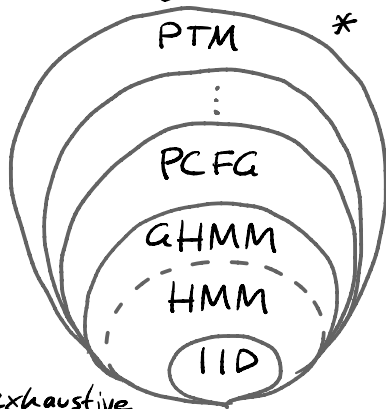
Progress can be broken down into two orthogonal directions:

- * For what classes of data generating processes do we understand optimal prediction?
- * For what scale of neural networks have we identified the structures of optimal prediction?

Processes:

Fix a (computable) stochastic process and consider:
What resources are required to generate this process?

This naturally leads to a hierarchy:



Probabilistic Turing Machine

Probabilistic Context-Free Grammar

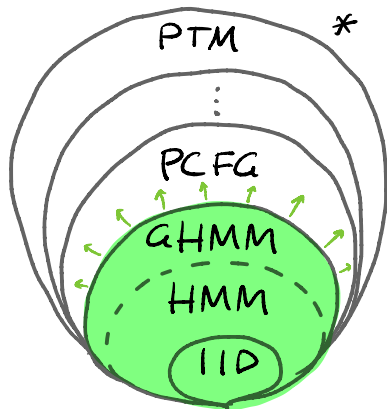
Generalised Hidden Markov Model

Hidden Markov Model

Indep. & Identically Dist.

Processes:

For what classes of data generating processes do we understand optimal prediction?



* not exhaustive

Probabilistic Turing Machine

Probabilistic Context-Free Grammar

Generalised Hidden Markov Model

Hidden Markov Model

Indep. & Identically Dist.

Neural Networks:

For what scale of neural networks have we identified the structures of optimal prediction?

* Transformer: $\sim 10^5$ parameters

* LSTM: $\sim 10^5$ parameters

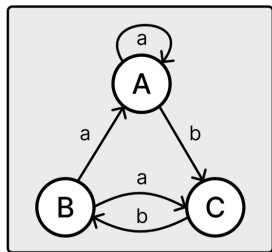
* RNN: $\sim 10^4$ parameters

* GRU: $\sim 10^5$ parameters

Very toy compared to current frontier models ($\sim 10^{12}$)!

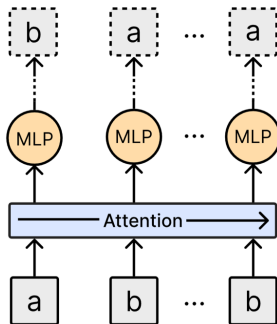
Example: Shai et al. 'Transformers represent belief state geometry in their residual stream'

HIDDEN MARKOV MODEL



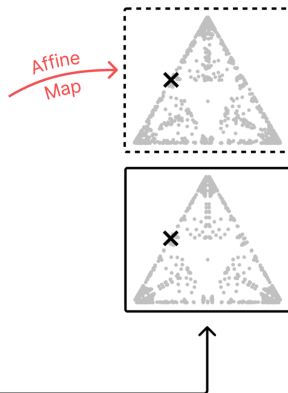
Emits Tokens

TRANSFORMER



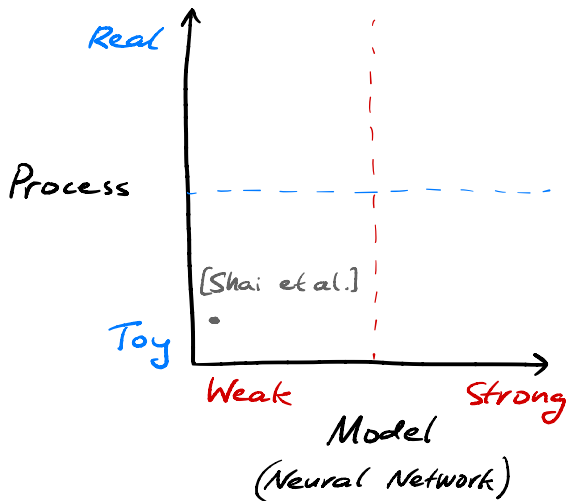
Bayesian Updating

BELIEF STATE



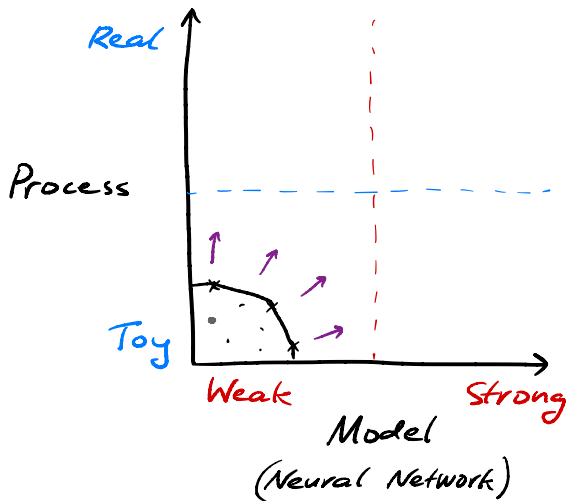
Overview - where is the program at?

We are working to push the Pareto frontier:



Overview - where is the program at?

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Overview - where is the program at?

We are working to push the Pareto frontier:

Wild Speculation:

As the research program is scale-free one view is that we should develop the foundations such that future automated AI Safety researchers can pursue these ideas independently

Weak

Strong

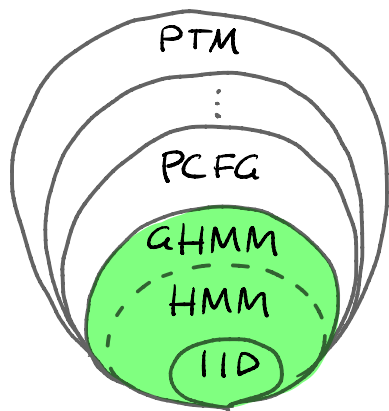
Model

(Neural Network)

Scope — what will we tackle today?

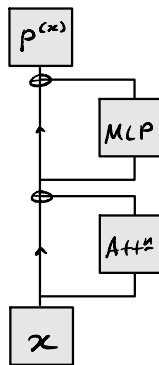
We will take a slightly narrower view and consider:

Processes



GHMMs

Models



Transformers

Plan:

- 1) Foundations - Hidden Markov Models [lecture/exercises]
- 2) Foundations - Prediction [lecture/exercises]
- 3) Applications - Belief geometry in transformers [reading]
- 4) Applications - Designing Processes [tutorial]

Questions?